

Advancing Critical Care Skills: Immersive VR Training Powered by Real-World Patient Data

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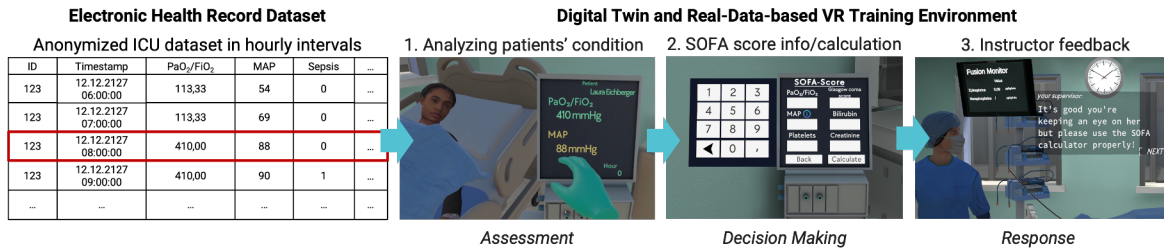


Figure 1: Flow of anonymized ICU data to the VR simulation, with hourly time advancement updating clinical parameters.

ABSTRACT

VR simulations are becoming essential for staff training in clinical care, specifically in environments that cannot be trained well during regular operation, such as critical care. This paper presents a novel system that aims at further closing the gap between simulation and reality by integrating real patient data into immersive training scenarios in the context of acute care. Focused on sepsis recognition, the VR simulation introduces trainees to clinical routines and to make informed decisions while observing the evolving patient conditions. By engaging with dynamic disease progression, it fosters understanding of critical conditions in a time-sensitive context. Preliminary feedback from a pilot assessment with nursing professionals highlighted its value for trainees and potential to enhance preparedness and decision-making skills in real-world scenarios.

Index Terms: Virtual reality, simulation, medical training.

1 INTRODUCTION

The shortage of workforce is a critical and pressing challenge for global healthcare systems, straining existing staff and limiting the quality of educational programs due to resource constraints [3]. Innovative solutions are required to bridge the gap between theoretical knowledge and practical experience. Virtual reality (VR) offers a platform for simulating realistic clinical environments where trainees can safely practice essential skills [2, 4]. We demonstrate a VR training application that is able to integrate real-world patient data, enhancing the realism of the experience. While traditional methods often present theoretical patient cases in a static manner, our simulation allows users to explore real-world cases and various stages of a patient's intensive care unit (ICU) stay, providing a dynamic view of disease progression. By engaging with authentic

clinical scenarios, users can apply knowledge in a realistic context, deepening their understanding of critical care. To demonstrate a first domain application, the prototype focuses on sepsis, a life-threatening condition that requires rapid recognition and action. The simulation trains users to apply the Sequential Organ Failure Assessment (SOFA) score, a key tool for assessing organ failure and predicting mortality in sepsis patients through data-driven interactive scenarios that reinforce informed clinical decision-making.

2 RELATED WORK

VR has proven effective in nursing education, particularly for teaching procedural techniques and emergency responses [2]. In the context of sepsis, various applications have been developed, such as an interdisciplinary VR training simulation that improved team-based decision-making in sepsis management [5]. In a PC-based multi-user simulation, Liaw et al. [1] found AI-based virtual doctors comparable to human-controlled ones in sepsis care and interprofessional communication, supporting their scalability for training. Advancing from existing approaches that are focused on knowledge transfer, our system aims at bridging the gap between theoretical learning and real-world scenarios and enhances critical thinking, decision-making, and adaptability in high-stakes situations.

3 METHODS

3.1 Virtual Environment

The system was developed using Unity3D 2021.3.12f1. Users are immersed in the role of a trainee nurse tasked with monitoring a patient in a ICU room. The environment and rooms are modeled on the basis of a real-world ICU. Users engage and interact directly with key clinical tools modeled after real-world workflows such as patient monitor, infusion pump, lab results monitor, and clipboard (Fig. 2). This interactive environment allows users to practice gathering and interpreting critical patient data, fostering practical decision-making skills that are difficult to replicate in traditional learning settings.

3.2 Scenario

The simulation starts with a guided tutorial introducing clinical parameters and critical thresholds. Users navigate the virtual room

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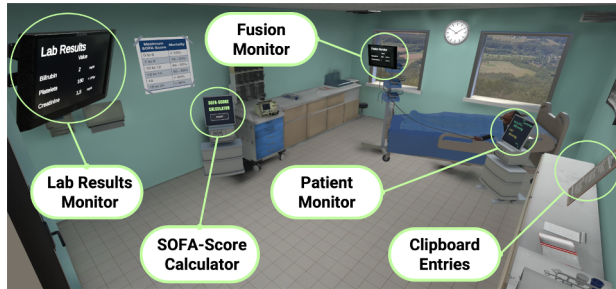


Figure 2: Overview of the simulated patient room: Users gather the relevant clinical parameters from the medical devices. The SOFA score is calculated using the built-in tool, allowing users to respond to the patient's condition as it evolves over time.

using teleportation to collect data and calculate the SOFA score using an integrated tool. The temporal progression of the patient's condition is defined by pre-fixed hourly intervals, representing the ICU timeline. However, the simulation allows users to progress at their own pace if they assess the patient as stable and decide to leave the room, which advances the simulation by one hour. If the assessment of sepsis is understood and the trainees have clear indications they can escalate concerns by contacting the supervisor via a phone call, mirroring real-world clinical workflows. Following such a scenario, they receive scripted feedback from a supervisor portrayed by a non-playable doctor, delivered through text bubbles, which assesses their clinical decision-making skills and reinforces best practices for managing critical conditions.

3.3 Data Integration

Patient data is sourced from a publicly available, anonymized ICU database, focusing on the eight critical values required for SOFA score calculation. This data is segmented into hourly intervals, enabling users to select different patients and experience diverse clinical scenarios based on real-world cases. The system supports dynamic integration of new datasets by placing properly formatted CSV files into a designated directory. A dataloader automatically detects and incorporates them during runtime, with clinical parameters displayed on virtual medical devices, mimicking real-world setups. Figure 1 describes the data integration and process. Users can start the simulation at various points in a patient's stay, offering insights into both current conditions and disease progression. When users determine that the SOFA score is not critical, the simulation advances by one hour. Upon their return, the clinical parameters are updated to reflect the patient's evolving condition. This progression allows users to observe how critical, time-sensitive conditions develop over time.

3.4 Exploratory User Study

The preliminary study involved five nursing and nursing education professionals (4 female, 1 male; ages 28-50, $M = 39.6$, $SD = 7.9$) with varying levels of professional experience ($M = 19.4$ years, $SD = 10.7$). One participant had used VR five times, the others had minimal to no experience. After completing the tutorial, participants gave qualitative feedback on the simulation's strengths, weaknesses, and the integration of real patient data, and rated the usefulness and the clarity of patient data integration on a five-point Likert scale.

4 PRELIMINARY RESULTS

Four participants found the tutorial easy to follow, with clear explanations and described the VR environment as intuitive. The simulation scenario was seen as closely aligned with reality, with

one participant noting its adaptive work environment and learning based on case studies. Two participants commented on the limited interactivity, noted the absence of patient communication and a more comprehensive supervisor feedback. While the simulation was beneficial for novice learners, experienced users found it partially predictable, with limited opportunities for independent decision-making. Participants rated the integration of real-world patient data as highly useful for improving their ability to interpret such data in everyday clinical practice ($M = 4.20$, $SD = 1.30$) and they found it well integrated into the simulation ($M = 4.60$, $SD = 0.55$). The importance of this feature for trainees was emphasized, with participants mentioning it would prepare them for real-life situations, potentially reducing feelings of being overwhelmed.

5 DISCUSSION

The simulation addresses key gaps in current training methods by combining real patient data with an immersive VR environment. Participants appreciated the balance between realistic scenarios and ease of use, which made the simulation accessible and practical. The integration of real clinical data not only enhanced the authenticity of the training but also fostered a deeper understanding of sepsis-related metrics. The study revealed further opportunities for refinement, such as introducing more dynamic interactions, including patient engagement and detailed supervisor guidance.

6 OUTLOOK

Future work will focus on integrating the system into a digital twin of an ICU, offering a highly realistic, dynamic training environment that simulates actual patient scenarios. This will enable advanced, real-time training and in-silico testing in otherwise difficult-to-access settings. Such a system will offer the opportunity to test various scenarios and decision-making strategies, making it a powerful tool for training in complex, high-pressure environments.

7 CONCLUSION

By integrating real patient data into our VR-based training environment for nursing trainees, the prototype provides practical insights into the temporal progression of critical conditions like sepsis, strengthening clinical decision-making in a realistic, data-driven setting. Despite minor areas for improvement, the prototype shows potential to become a valuable tool for training novice healthcare professionals, offering a structured introduction to ICU workflows, decision-making and improving patient outcomes.

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